

Optimization of airfoil Based Savonius wind turbine using coupled discrete vortex method and salp swarm algorithm

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ABSTRACT

In this research study, the optimization study of an airfoil type Savonius turbine is presented, focusing on maximizing the power coefficient. For this purpose, a Discrete Vortex Method (DVM) code for calculating power coefficient, a Class Shape Transfer (CST) function code for generating coordinates of airfoils and a Salp Swarm Algorithm (SSA) code for optimization have been developed. These three codes are coupled and an optimization procedure is generated. The decision variables of the mentioned problem are the CST function coefficients and the objective is the power coefficient. For verification of the results, CFD simulations are utilized. The results indicate that the achieved geometry could increase the power coefficient of a conventional Savonius turbine up to 27%. Also, the physical results indicate that the suction area and pressure magnitude of the pressure side of the airfoil type turbine is more extensive than simple type Savonius turbine at the optimization TSR. Therefore, in the airfoil type wind turbine, the great pressure difference causes more force at the surface of blades and in results, an increase in power coefficient occurs.

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1. Introduction

In recent years, for many reasons, including fossil fuels reduction, environmental pollution, and global warming, energy production from renewable sources is a priority of the international community. Among different renewable energy types, which include solar, wind, hydro, and so forth. Wind energy has been studied for reasons such as being clear and abundant. Primarily, there are two categories of wind turbines: the horizontal axis and the vertical axis wind turbines. Different types of vertical axis wind turbines exist, but the two major groups are Savonius (Drag-type) and Darrieus (Lift-type) turbines. Generally, vertical axis wind turbines have advantages over horizontal axis wind turbines because of their small tower sway, their independence to wind direction, their less noise production, being less obstruction for birds, maintenance costs, etc. Both Savonius and Darrieus turbines have their own advantages and disadvantages. The advantage of Darrieus rotors is their higher efficiency in comparison to Savonius turbines (The maximum power coefficient of Savonius turbines depending on their design and operation condition can vary

between 0.04 and 0.2 (Aslam Bhutta et al., 2012; Gorelov and Krivospitsky, 2008; Wenehenubun et al., 2015; Mahmoud et al., 2012; Shaughnessy and Probert, 1992; Montelpare et al., 2018; Tahani et al., 2017)) while the most significant advantage of Savonius turbines are characterized by their ability to self-start, meaning they can produce higher starting torques in comparison to Darrieus rotors. These features of both turbines make them very suitable to be combined with each other and produce higher power coefficients. According to the mentioned point, studies on the Savonius turbines are essential because they can be an assistant to the Darrieus turbines and also they can be used at commercial scale for power generation by optimizing their geometry. The focus of the present research study is on the Savonius type of wind turbines.

According to the available literature about wind turbine, many types of researches including experimental investigations, numerical simulations, and design optimization are conducted that some of them are presented here (Zhang et al., 2018; Huet al, 2018; Wang and Teah, 2017; Ayodele et al., 2016; Novaes Menezes et al., 2018; Niu et al., 2018). Kamoji et al. (2009) investigated conventional Savonius or modified forms of the conventional Savonius rotors to obtain a uniform coefficient of static torque. Roy and Saha (2013) considered a newly developed two-bladed Savonius-style wind turbine and studied the effects of Reynolds number on the dynamic and static characteristics of the turbine. Golecha et al. (2011)

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Nomenclature			
A	Curvature coefficients	S	Shape function
ALE	Leading edge of A	$\vec{U}_0$	Free stream velocity (m/s)
ATE	Trailing edge of A	W	Complex velocity (m/s)
BEM	Blade Element Momentum	e	Spacing between two airfoils
BLE	Leading edge of B	$\vec{r}$	Position Vector
BTE	Trailing edge of B	$\vec{s}$	Tangential unit vector
CFD	Computational Fluid Dynamics	TSR	Tip Speed Ratio
C	Class function	t	Time instant
C_M	Torque coefficient	TKE	Turbulence Kinetic Energy
C_P	Power coefficient	ub	Upper bound
CST	Class Shape Transfer	z	Coordinates of a point in the complex plane
DVM	Discrete Vortex Method	<i>Greek letters</i>	
$\vec{F}$	Aerodynamic force	α	Angular position
FVM	Finite Volume Method	γ	Vortex intensity (m ² /s)
L	Maximum number of iterations in SSA	ρ	Density (kg/m ³)
$\vec{M}$	Torque (N.m)	σ	Limit spacing between two free vortices
PSO	Particle Swarm Optimization	ϕ	Potential function
P	Power (W)	ψ	Stream function/the ratio of x/c in CST
R	Rotor radius (m)	$\vec{\omega}$	Angular velocity (rad/s)
SSA	Salp Swarm Algorithm		

conducted a study to find out the optimal position of the deflector plate in a Savonius wind turbine. Behrouzi et al. (2019) suggested a novel drag based vertical axis current turbine using arms and self-adjusting blades and the results showed there was a 30.7% increase in performance for the self-adjusting blade turbine. Ogawa (1984) in 1984 for the first time modeled the Savonius wind turbine with discrete vortex method (DVM) and the results obtained by the DVM did not agree with the experimental data quantitatively but agreed well in tendency. Fernando and Modi (1989) investigated the effect of viscosity in the Savonius with DVM and they achieved that Savonius rotor is not a pure drag device and is also affected by lift. Kotb and Aldoss (1991) investigated the effects of adding a duct to the Savonius by using DVM. Fujisawa (1996) studied Savonius, experimentally and numerically (DVM) and examined the overlap ratio effect on the flow field and founded that the performance could be improved with small overlap ratios. Later, Afungchui et al. (2014) studied the wake of Savonius with DVM and the results were in good agreement with the experimental visualizations. The latest use of the discrete vortex method in the Savonius wind turbine was in 2016 by Wang and Yeung (2016) and the study was about the evaluation of the suitability of Bach-type turbines as micro-scale energy devices. According to the presented literature, DVM has been widely used in modeling of Savonius wind turbines because of its simplicity in comparison to the other numerical methods. By following the mentioned point, in this research study, DVM has been used for modeling of Savonius type wind turbine at the initial stage.

Optimization has a great impact on all fields of science and engineering. Nowadays, all researches are mostly looking for algorithms which are gradient-free, robust and straightforward. Nature inspired optimization algorithms have the mentioned properties and have been widely utilized in the field of wind engineering for harnessing the maximum amount of wind flow kinetic energy. For example, Tahani et al. (Tahani et al., 2017) used ant colony optimization algorithm, blade element momentum (BEM) theory, and CFD to optimize the geometry of a horizontal axis wind turbine; the results indicated that the achieved geometry by the optimization has a power coefficient 13.7% greater than the base

geometry. Later, Tahani et al. (2016) studied about a novel heuristic method for optimizing the vertical axis wind turbine design and achieved a power coefficient, 44% higher than the maximum power coefficient of the original turbine. Mohamed et al. (2011) optimized the Savonius wind turbine by coupling CFD and C program and found that the performance increase exceeds 30%. Tian et al. (2018) optimized the blade in a Savonius wind turbine with PSO algorithm and the results showed that the achieved blade could produce a power coefficient, 4.41% higher than the traditional design. Beside nature-inspired optimization algorithms, several researchers have optimized the performance of wind turbines, using different novel ideas or methods. For example, Milas et al. (Marinić-Kragić et al., 2018a) caused a 15% increment in power coefficient by attention to shape parameterization scheme based on B-splines. Shabbir et al. (Moazam Sheikh et al., 2017) studied about parametric optimization of a Savonius wind turbine, using response surface methodology and the effects of shape factor, overlap ratio, and tip speed ratio on turbine performance was studied. Mari et al. (2017) studied new rotor geometry with spline function and the results indicated that the power coefficient was increased by 20%. In the research study carried out by Sharma and Sharma (2017), the performance of a Savonius rotor has been improved using multiple quarter blades. According to the results, the performance of the turbine improved by 8.89–13.69% in comparison to previously published studies. Recently, Roy et al. (Roy and Ducoin, 2016) proposed an inverse method for optimization of Savonius turbines. In this research study, the proposed inverse technique uses a differential evolution algorithm for optimization of the size and also the space of the turbine. Chan et al. (2018) used genetic algorithm to optimized the conventional Savonius wind turbine and succeeded in improving the power coefficient at TSR equal to 0.8. Zhou et al. (2018) studied the geometry optimization based on two-dimensional discrete cosine transformation (2D-DCT) through numerical simulations and the proposed technique enhanced the efficiency by 13.77% at TSR equal to 1.0 and 21.11% at TSR 0.0838. Ostos et al. (Iván et al., 2019) tried to advance the performance of a Savonius wind turbine by introducing a new configuration without expensive and complex modifications. They could increase the performance of the

wind turbine with 17.81%. Marinić-Kragić et al. (Marinić-Kragić et al., 2018b) investigated a novel VAWT idea which is a Savonius wind turbine with flexible blades that change their shape passively due to aerodynamic and blade inertial forces during rotation. This optimization was based on genetic algorithms. The results show that 8% improvement in the power coefficient is possible.

Advantages of the Savonius wind turbine such as self-starting ability at low wind speed, independence to wind direction and low cost have made these rotors suitable for urban areas. The performance of these turbines can significantly influence economic justification. According to the presented literature, several studies have been carried out for the optimization of the Savonius turbines. The primary aim of this research study is optimizing the performance of these types of rotors. Commonly, there are several types of optimization procedures for the Savonius turbine. Using a novel specific geometry or changing the geometrical parameters of the conventional models using CFD solver, are popular procedures that are investigated in previously published studies (Niu et al., 2018) (Behrouzi et al., 2019). As was mentioned earlier, one of the main findings of the Fernando and Modi (1989) was that these types of turbines are not pure drag based and lift forces also affect the performance of the turbine. Based on this finding, the idea of utilization of airfoils has been investigated in the present research study. A similar study has been previously carried out by Tartuferi et al. (2015) in which the application of two different airfoils has been investigated. The presented study has the following differences with the previously published research study by Tartuferi et al. (2015):

- Developing the unsteady discrete vortex method code to calculate the power coefficient of the turbine.
- Developing a class shape transfer (CST) function method code, to express the coordinates of the airfoil as a polynomial function and decrease the complexity of the optimization process in order to make the optimization procedure easier.
- Developing the salp swarm algorithm (SSA) code for the purpose of finding the best curvature coefficients of the best airfoil that produces the maximum amount of the power coefficient.
- Coupling the CST, DVM and SSA codes, to obtain the optimum power coefficient and optimum airfoil design for Savonius type turbine. Most of the previously published research studies have used a CFD solver beside an optimization code while in this research study a package of DVM/CST/SSA code has been developed.
- Obtaining the optimum airfoil coordinates for the Savonius turbine according to the defined boundaries.
- Verification of the achieved results by CFD simulation.

In summary, improving the performance of Savonius turbine using the above-mentioned methodology is the primary goal of this study. As was mentioned earlier, by optimization of these types of turbines with using novel and optimum blade geometries, their application domain and also the attraction of investors and researchers to these types of turbines will be increased. The presented paper is organized as follow. First, in section 2, the methodology, containing the DVM, CST, SSA and optimization procedure are explained. In section 3, the results of optimization are presented and CFD simulations are discussed. In section 4, the main findings of papers are concluded.

2. Methodology

In this section, the method of solution which is the discrete vortex method is presented and the optimization algorithm is described. In the end, the process of coupling the solution and optimization methods are explained.

2.1. Discrete vortex method

The modeling of flow is carried out using the discrete vortex method which is a subset of a singular element based numerical solution and the geometry has been discretized to the panels by putting the vortex in the middle of the panel.

The DVM method has several advantages like solving fluid dynamics problem in a fast way in comparison with the Finite Volume Method (FVM). Also, compared with FVM the implementing of this method is clear and easy to use and there is no need for creating the complicated grid in the DVM method. However, this method has some limitation and assumptions which must be considered for a reliable simulation. The basic assumptions for the flow field in this method is the inviscid, irrotational, incompressible, unsteady flow with no body forces. The drag force is formed from two part, the viscous drag and pressure drag. The viscous drag is created due to the viscosity and the interaction between the fluid and solid, but the pressure drag is formed because of the difference between flow pressure in a wind turbine. According to the reference (D'Alessandro et al., 2019), in a wind turbine, the main part of the drag force is created by pressure drag. Therefore, the inviscid assumption is acceptable for the purpose of modeling the Savonius wind turbine.

In this section, it is attempted to demonstrate the DVM, but more information can be found in reference (Katz and Plotkin, 2001).

2.1.1. Discretization of the airfoil

In the present study, the two same airfoils create the geometry of the Savonius wind turbine cross section. The rotor diameter and overlap ratio are the fundamental parameters in the Savonius rotor. A schematic of the airfoil based Savonius turbine has been shown in Fig. 1. As can be seen, e is the length of overlap of two airfoils and H is the height of the wind turbine.

In this work, three coordinate systems are used. First, the fixed frame is shown by (X, Y, Z) . Second, The body frame is indicated by (x, y, z) . Third, the unit vectors which are tangential and normal at each point, are represented by (n, s) . By rotating the Savonius rotor, an angle (α) is being created between the body frame and the fixed frame, which is related to the angular velocity (ω) of the rotor and time (t) .

According to Fig. 2, each airfoil is divided into n panels, then bound vortex is distributed in the middle of the panels. Control points are placed at the beginning and the end of each panel (except the first panel). The free vortices are emitted from the

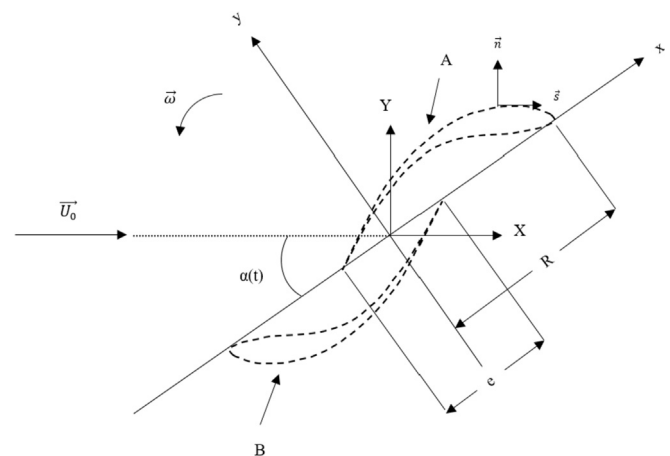


Fig. 1. Airfoil coordinates.

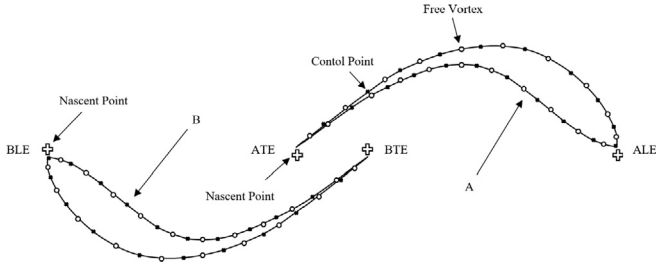


Fig. 2. Discretization of the airfoils.

leading edge (LE) and trailing edge (TE). N bound vortex and two free vortices for each airfoil are unknown therefore $2n + 4$ equations are necessary for modeling the flow; $2n - 2$ equations from applying the boundary conditions at control points are available and two equations from Kelvin theorem (Katz and Plotkin, 2001) are obtained and 4 equations are achieved from Kutta condition (Katz and Plotkin, 2001).

2.1.2. Unsteady flow model

The upcoming wind velocity is $\vec{U}_0$ and the rotor is rotating with angular speed ω . The velocity can be derived from complex potential which can be found at a point z from eq. (1) (Fernando and Modi, 1989).

$$W_i(z) = \frac{I}{2\pi} \gamma_i \ln(z - z_i) \quad (1)$$

In Eq. (1), W is equal to $\phi + i\psi$ (ϕ and ψ are potential and stream functions, respectively) and the coordinate of each point is presented by z and finally, the strength of the vortex has been shown by γ .

The flow tangency condition is being applied to the $2n - 2$ control points. This condition says that the normal component of velocity at the surface of the body is zero. By using eq. (1) and considering that λ represents one of the airfoils (A or B), the flow tangency at any control point (z_i) is given by:

$$\left[\vec{U}_0 - (\vec{\omega} \times \vec{r}_i) + \left(\sum_p \frac{I(\gamma_A)_p}{2\pi(z_i - (z_A)_p)} + \sum_p \frac{I(\gamma_B)_p}{2\pi(z_i - (z_B)_p)} \right) + \left(\sum_{q=1}^2 \frac{I(\gamma_{AE})_q}{2\pi(z_i - (z_{AE})_q)} + \sum_{q=1}^2 \frac{I(\gamma_{BE})_q}{2\pi(z_i - (z_{BE})_q)} \right) + \left(\sum_{p=1}^{k-1} \frac{I(\gamma_{WAE})_{pq}}{2\pi(z_i - (z_{WAE})_{pq})} + \sum_{p=1}^{k-1} \sum_{q=1}^2 \frac{I(\gamma_{WBE})_{pq}}{2\pi(z_i - (z_{WBE})_{pq})} \right) \right] \cdot \vec{n}_{\lambda i} = 0 \quad (2)$$

Where the $\vec{n}_i$ is the normal direction at any given point and time iteration is represented by k . The second boundary condition is Kutta–Joukowski condition. Because of two emission points in the airfoil, two Kutta conditions are required which are applied at trailing edge and leading edge. For this purpose, the sum of the two vortices at the end of the leading edge and trailing edge must be zero (Katz and Plotkin, 2001). By applying the Kelvin's theorem to the airfoils, the last two equations are achieved. These are given by:

$$\sum_p (\gamma_A)_p^t + \gamma_{ATE}^t + \gamma_{ALE}^t = \sum_p (\gamma_A)_p^{t-1} \quad (3)$$

$$\sum_p (\gamma_B)_p^t + \gamma_{BTE}^t + \gamma_{BLE}^t = \sum_p (\gamma_B)_p^{t-1} \quad (4)$$

2.1.3. Numerical calculation

If the spacing between the two vortices is small, the velocity becomes unrealistically large. To avoid this, the way to calculate the speed is changed. If the distance between the two vortices is smaller than the $\sigma = 0.05(\text{Chord})$, the velocity is being calculated from $(z) = \frac{I\gamma_i}{2\pi\sigma^2} (\bar{z} - z_i)$. Also if the space between vortex and body is smaller than the σ , the vortex is removed for the effect of viscosity. Roll up is the method that is used for the modeling of the vortex shedding in this study. This technique is the moving of the vortices with the speed that is induced by other vortices in the last iteration.

2.1.4. Torque and power coefficient

The pressures can be computed by using the Bernoulli equation near the panel (Katz and Plotkin, 2001):

$$\frac{p_{ref} - p}{\rho} = \frac{U^2}{2} - \frac{U_{ref}^2}{2} + \frac{\partial \phi}{\partial t} \quad (5)$$

From Eq. (6) (Katz and Plotkin, 2001) the pressure difference between the upper and lower curves at each panel can be derived.

$$\Delta p_i = p_u - p_l = \rho \left[\frac{U_{ref}^2}{2} - \frac{((\vec{U}_0 - \vec{\omega} \times \vec{r}_i + \vec{u}_{wake}) \cdot \vec{\tau}_i + \frac{\gamma_i}{2l_i})^2}{2} - 0.5 \frac{\delta}{\delta t} \sum_{k=1}^i \gamma_i \right] \quad (6)$$

The unsteady aerodynamic force is obtained by the following equation:

$$\vec{F}_i = \Delta p_i \times \Delta S_i \times \vec{n}_i \quad (7)$$

The torque can be calculated from Eq. (8):

$$\vec{M} = \sum_{i \in A+B} (\vec{r}_i \times \vec{F}_i) \cdot \vec{k} \quad (8)$$

Finally, the torque coefficient, power and power coefficient can be found by using the following equations:

$$C_M = \frac{|\vec{M}|}{0.5 \rho S C U_0^2} \quad (9)$$

$$P = |\vec{M}| \omega \quad (10)$$

$$C_P = \frac{P}{0.5 \rho S U_0^3} \quad (11)$$

2.1.5. Validation of developed code

In Table (1) a comparison between the results of the developed discrete vortex method and experimental measurements

conducted by Sheldahl et al. (1978) in a varied range of TSR has been presented.

The difference between the numerical and experimental results is due to the assumptions made in the problem such as inviscid, irrotational, 2D, etc. With considering the trend of power coefficients which is similar to the experimental results and the fact that according to the references (Ogawa, 1984), (Fernando and Modi, 1989), (Afungchui et al., 2009) this error is logical and acceptable for this method; the error which we have reached is reasonable.

2.2. Class shape transfer (CST) function method

In airfoil optimization, if each point of pressure and suction surfaces was considered as a decision variable, the complexity of the optimization process would be increased and also several constraints must be defined to achieve a smooth geometry. Therefore, in this research study, the CST method has been used to decrease the mentioned complexities. CST parameterization technique is a potent tool for airfoil smoothing. According to this method, the upper and lower surfaces of an airfoil are defined as follow (Lane and Marshall, 2010), (Kulfan and Hilton, 2007):

$$\zeta_U(\psi) = C_{N_2}^{N_1}(\psi) \cdot S_U(\psi) + \psi \cdot \Delta \zeta_U \quad (12)$$

$$\zeta_L(\psi) = C_{N_2}^{N_1}(\psi) \cdot S_L(\psi) + \psi \cdot \Delta \zeta_L \quad (13)$$

Where ψ is the x/c and ζ is the z/c . The subscripts U and L present the upper and lower surfaces. The class function C is being presented as follow (Lane and Marshall, 2010), (Kulfan and Hilton, 2007):

$$C_{N_2}^{N_1}(\psi) = \psi^{N_1} \cdot (1 - \psi)^{N_2} \quad (14)$$

For round nose and pointed aft end airfoils, N_1 is 0.5 and N_2 is 1. In equations (12) and (13), the S presents the shape function and is expressed as below (Lane and Marshall, 2010), (Kulfan and Hilton, 2007):

$$S_U(\psi) = \sum_{i=0}^{N_U} A_U(i) \cdot S(\psi, i) \quad (15)$$

$$S_L(\psi) = \sum_{i=0}^{N_L} A_L(i) \cdot S(\psi, i) \quad (16)$$

In the above equations, A is a set of curvature coefficients by which the airfoil can be presented and N is the order of the Bernstein polynomial. $S(\psi, i)$ is being presented using a Bernstein polynomial as follow (Lane and Marshall, 2010), (Kulfan and Hilton, 2007):

$$S(\psi, i) = K_i^N \cdot \psi^i \cdot (1 - \psi)^{N-i} \quad (17)$$

Table 1
The difference between developed DVM code and experimental results extracted from (Sheldahl et al., 1978).

TSR	0.35	0.45	0.58	0.62
Numerical	0.17	0.22	0.255	0.27
Experimental	0.14	0.18	0.21	0.22
Error	21%	22%	21%	23%

$$K_i^N = \frac{N!}{i!(N-i)!} \quad (18)$$

By combining equations (12)–(18), the complete curvature of upper and lower surfaces of the airfoil can be presented as follow:

$$\zeta_U(\psi) = \psi^{0.5} \cdot (1 - \psi)^{1.0} \sum_{i=0}^{N_U} \left[A_U(i) \cdot \frac{N_U!}{i!(N_U-i)!} \cdot \psi^i \cdot (1 - \psi)^{N_U-i} \right] + \psi \cdot \Delta \zeta_U \quad (19)$$

$$\zeta_L(\psi) = \psi^{0.5} \cdot (1 - \psi)^{1.0} \sum_{i=0}^{N_L} \left[A_L(i) \cdot \frac{N_L!}{i!(N_L-i)!} \cdot \psi^i \cdot (1 - \psi)^{N_L-i} \right] + \psi \cdot \Delta \zeta_L \quad (20)$$

According to the above equations, the shape of the airfoil depends on the A_U and A_L . Therefore, the set of these coefficients can be considered as decision variables for optimization. The complete explanation of the CST method can be found in (Lane and Marshall, 2010)–(Kulfan, 2007).

2.3. Salp swarm algorithm (SSA)

In recent years, the application of nature-inspired optimization algorithms has been increased in several fields of science, engineering, and technology. This is because these algorithms are gradient-free, they avoid local optimums and are very suitable for handling complex optimization problems. These algorithms must have three characteristics. The first one is global search or exploration (diversification). In the global search phase, the algorithm must search the entire search space efficiently. The second one is the local search or exploitation (intensification). In this phase, the algorithm must search locally and must search intensified around the optimum solutions which have been found so far. A too much global search will make the algorithm to converge too late meaning that it will cause a delay in convergence. On the other hand, too much local search will cause the trapping of the algorithm. Therefore, a proper balance must be created between these two characteristics. The elitism is the mechanism of the algorithm for selecting good solutions and sends them to the next iteration. The operating mechanism of each meta heuristic algorithm is as follow:

- First, a population of random solutions are being generated and then their objective function is being calculated.
- Then all solutions are being sorted according to the value of their objective function and the best solution is being found.
- In the next step, the algorithm enters its main loop which can be a fixed number of iterations or it can be a criterion.
- At each iteration, by using the algorithm equations, new solutions are being generated.
- Then the objective function of each solution is being calculated.
- According to the elitism mechanism, by a combination of previous solutions and new solutions, a new set of solutions are being selected to be transferred to the next iteration.
- By continuing this procedure, the algorithm approaches the optimum solution and finally, it converges to the global best solution.

Several algorithms have been proposed so far and the purpose of all researches is to develop robust algorithms with less dependent parameters. In the latest algorithms, the convergence of the algorithm is less dependent on the number of populations, but it is

worth mentioning that a quick sensitivity study would help the researcher in selecting the number of populations. Because in some cases, such as the problem which is presented in this study, the required run time is high, therefore it is recommended to select the number of populations as small as possible. In recently developed algorithms, the purpose is increasing the simplicity of the algorithm and also at the meantime the robustness of it against different types of problems. Also decreasing the user-defined parameters is one of the important targets of novel algorithms. Salp Swarm Algorithm (SSA) is one of the recently proposed methods by Mirjalili et al. (2017). Inspired by the chaining behavior of salp swarm. In this algorithm, the population of salp is being divided into two parts. The first part has only one member which is the swarm leader and the second part are the followers. At each iteration, the position of the leader is being updated according to the following equation (Mirjalili et al., 2017):

$$x_j^1 = \begin{cases} F_j + c_1((ub_j - lb_j)c_2 + lb_j) & c_3 \geq 0 \\ F_j - c_1((ub_j - lb_j)c_2 + lb_j) & c_3 < 0 \end{cases} \quad (21)$$

Where, x_j^1 is the position of the j th dimension of the leader, F_j is the position of the food, that for simplicity can be considered the global best solution. ub_j and lb_j are the upper and lower bounds of j th dimension respectively. c_2 and c_3 are random numbers. For the purpose of creating balance between local and global searches, c_1 is being defined as follow (Mirjalili et al., 2017):

$$c_1 = 2e^{-(\frac{4}{t})^2} \quad (22)$$

The position of followers is being updated according to Newton's law of motion and can be expressed as follow (Mirjalili et al., 2017):

$$x_j^i = \frac{1}{2}(x_j^i + x_j^{i-1}) \quad (23)$$

The detail explanation of SSA can be found in (Mirjalili et al., 2017).

2.4. The optimization procedure

As was mentioned earlier, in this research study, the purpose is to develop an optimized geometry for an airfoil-shaped Savonius type wind turbine. For this purpose, the salp swarm algorithm has been coupled with the vortex method. The CST method curvature parameters are considered as the decision parameters and the power coefficient has been considered as the objective function. In all nature-inspired optimization algorithms, the first step is to generate a population of variables between the upper and lower bounds of each variable. In this research study, because the variables are the coefficients of a polynomial, therefore, the upper and lower bounds of curvatures must be first selected. Fig. 3 illustrates the upper and lower bounds of the coordinates. These boundaries are selected according to the available literature and also physical observations. Airfoils such as CH10 smoothed, Chen airfoil, E 216, Eppler 387, Eppler 58, FX 082-512, SR 3345, SR 5050 and etc. have a high lift. In addition, according to the fact that the simple Savonius turbine is a drag-based turbine, therefore the coordinates of the pressure surface must definitely be positive. By combining this point and the coordinates of the high lift airfoils, the upper bounds of the curvature coefficients of the suction surface were selected in a manner that the generated curvature covers the high lifts airfoils suction surface coordinates. In addition, the lower bounds of the curvature coefficients of the pressure surface were selected according to the same logic and satisfying the positive coordinates constraint. The upper bound of the pressure surface and lower

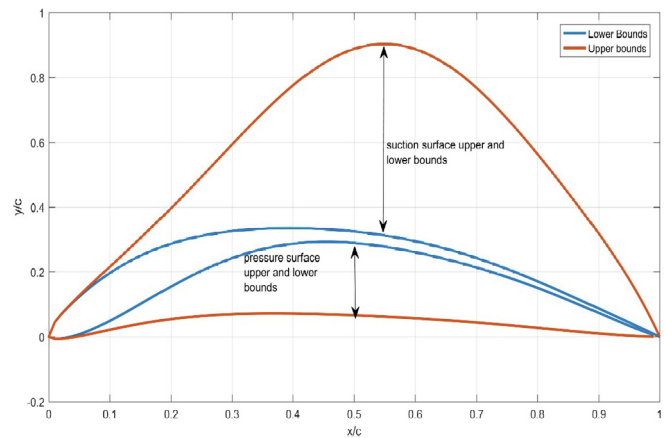


Fig. 3. Upper and lower bounds of the coordinates.

bound of the suction surface are considered the coordinates of SR 3345. This decision was based on the published results of Tartuferi et al. (Golecha et al., 2011), in which it is illustrated that the SR 3345 rotor produces a higher power coefficient at TSRs below 0.6.

The order of the Bernstein polynomial is considered equal to 6. The upper and lower bounds of the first coefficients of suction and pressure surfaces are considered equal. Therefore, in a presented research study, eight variables must be optimized to get the maximum power coefficient. The upper and lower bounds of each coefficient are mentioned in Table (2).

The optimization procedure is illustrated in Fig. 4. According to this figure, the main steps of the utilized algorithm is as follow. At the first step an initial population of the curvature coefficients is being generated and the proportional coordinate of each generated solution is being calculated according to CST method. After the mentioned step, the generated coordinates are entered into discrete vortex code and the power coefficient of each solution is being calculated. The next step is sorting solutions according to the value of their power coefficient and generating new solutions using SSA. The mentioned step is being repeated until a convergence criterion is being satisfied.

3. Results and discussion

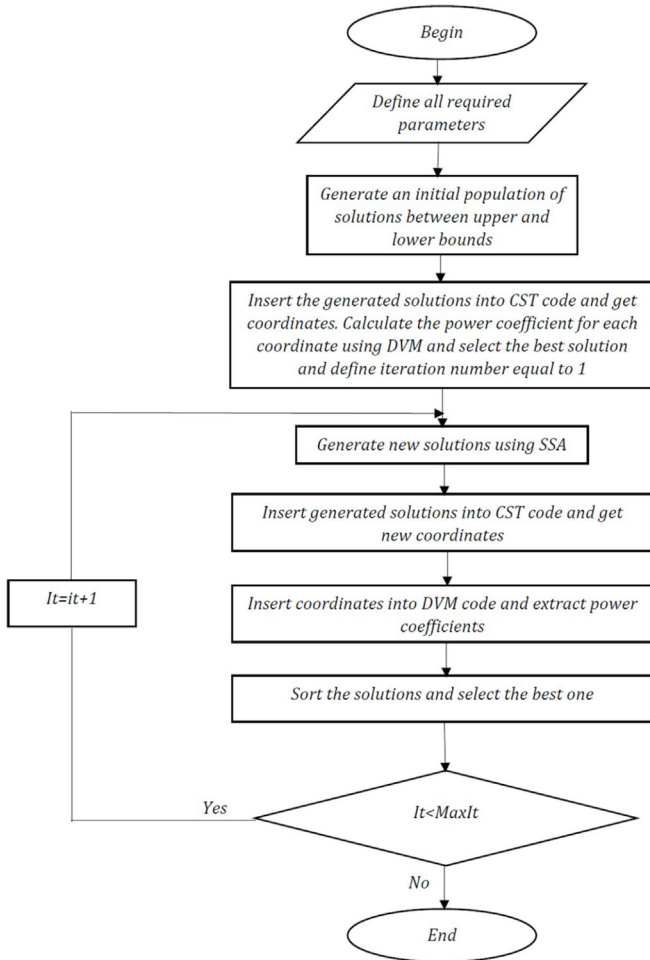
3.1. Optimization results

Table (3) illustrates the selected parameters of the algorithm and also a panel method code. As it can be seen the optimization has been carried out in TSR equal to 0.4 and free stream velocity equal to 7 m/s. As was mentioned, two types of scenarios can be used for stopping the algorithm from the processing: 1) fixed number of iterations, 2) convergence criterion. One of the main advantages of using a fixed number of iterations is that if the algorithm is being trapped in a local optimum, it will have the chance to skip from it. In this research study, the number of iterations was selected high enough in order to give the algorithm the chance of skipping from local optima. The convergence of the algorithm was starting much sooner. Regarding the number of populations, when the required processing time is high, it is necessary to select the number of populations as small as possible. Therefore, according to the number of population and also a quick parametric study, the number of populations was selected equal to 15. Fig. 5 illustrates the optimum geometry that can produce the maximum amount of power coefficient. The coordinates of the obtained geometry are presented by star marks. The presented optimum geometry can produce a power coefficient equal to 0.085. The performance

Table 2

Upper and lower bounds of the variables.

	A_u	A_u	A_u	A_u	A_u	A_u	A_l	A_l	A_l	A_l	A_l	A_l
Min	0.416	1.21	0.478	1.27	0.91	0.89	−0.086	0.356	0.11	0.198	0.234	0.026
Max	0.416	1.21	0.975	5.15	1.72	4.06	−0.086	0.356	1.40	0.365	1.25	0.553

**Fig. 4.** The optimization flowchart.**Table 3**

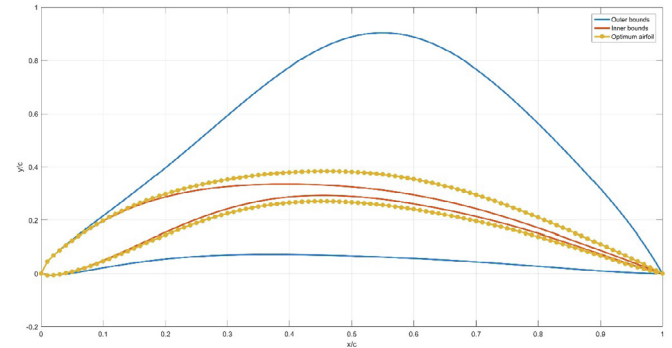
The inputs of the algorithm and also panel method code.

Inputs	
SSA parameters	Maximum Number of Iteration = 100 Number of Population = 15
DVM parameters	Space between two blades = 10cm Blades Chord = 50cm Tip Speed Ratio = 0.4 Turbine Height = 1m Free Stream Velocity = 7 m/s density = $1.2 \frac{\text{kg}}{\text{m}^3}$

comparison of the achieved geometry with a conventional Savonius turbine has been carried out in the next section.

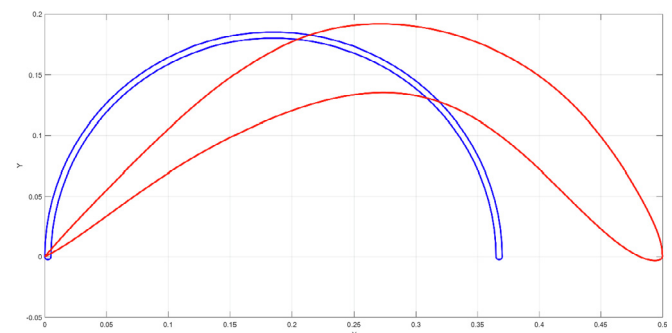
3.2. Computational fluid dynamics (CFD) verification

According to the existing error for the DVM, it was decided to

**Fig. 5.** The achieved optimum geometry coordinates.

verify the achieved results using the CFD method. The utilized methodology is used both for the semicircular Savonius turbine and also the airfoil type turbine. To achieve a reasonable physical comparison between semicircular and airfoil type turbines, it was assumed that the inner curve length of both turbines must be equal. Since the viscous drag is related to the length of the blade, it must be sure that this ignored part of the drag will be constant at each geometry in order to achieve a reasonable optimum geometry. Also, by this metric, the production cost can be compared in a way. If the production cost is related to the length of the blade, it can be assumed that an optimum geometry with a constant production cost has been achieved. As it is illustrated in Fig. 6, the generated geometry for the semicircular Savonius turbine is based on the length of the pressure side of the obtained airfoil. Both geometries have the same length of pressure side.

The utilized computational domain has been shown in Fig. 7. Also, for the modeling of turbulence, the $SSTk - \omega$ model is selected. According to the references (Roy and Saha, 2013), (Mari et al., 2017), (Tartuferi et al., 2015), (Wong et al., 2017), the 2D assumptions is valid because all of these references use a 2D domain for their simulation or explanation. This is owing to the fact that this kind of turbine does not have any change in the Z direction and there is no significant 3D phenomena in this type of wind turbine. Also, in many researches as well as that references, the researchers use their results in 2D for validation with the experimental result. This model was chosen because according to the available literature, it

**Fig. 6.** The Airfoil and the semicircular simple Savonius coordinate.

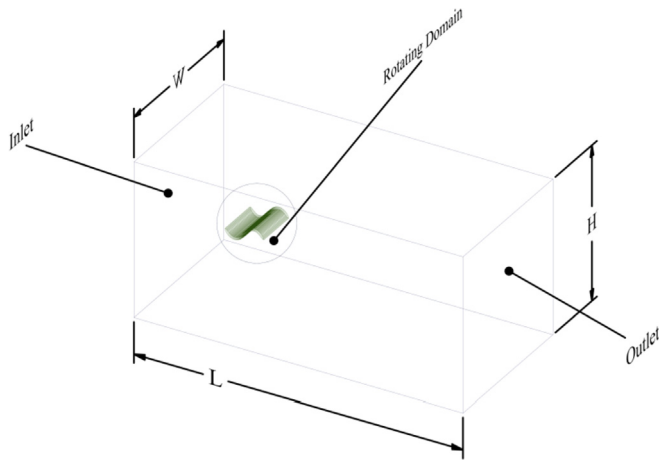


Fig. 7. The 3D Computational domain.

has a very good agreement with experimental data (Moazam Sheikh et al., 2017) (Tartuferi et al., 2015) (Roy and Saha, 2013). The following explained method is used for validating the CFD procedure at TSR equal to 0.45 and the achieved results indicated that a great agreement exists between them and experiment (Sheldahl et al., 1978). This is because the maximum error for the power coefficient is 3%.

3.2.1. The computational domain

The utilized computational domain contains two sub-regions. The first region is a sphere that rotates with constant rotational speed and having a diameter of $1.2D$ where D is the rotor diameter and is 0.95 m. The second region is a cube that is fixed and its dimensions are $W = 12D$, $L = 20D$, $H = 12D$. Three kinds of boundary condition (B.C.) is used. The B.C. for the inlet is velocity inlet. For outlet B.C. the pressure outlet is chosen and the magnitude of gage pressure is considered to be equal to zero. For the walls of the cube, the B.C. is considered as a wall. The walls of the cube could be wall or openings/outlets. The wall boundary condition has been used in references (Tian et al., 2018), (Tartuferi et al., 2015) and either of these options is correct. But when the wall boundary condition is selected, a far distance between the blade and the walls must be created in order to avoid any possible influence of wall in fluid flow. Consequently, a long distance between the blades and walls has been selected.

Since in this work $SSTk - \omega$ model has been used for simulating the turbulence, y^+ must be less than 5 (Fluent Inc. User's Guide, 2018). In order to have a good quality and resolution in the boundary layer grid, the y^+ has been considered equal to 1. The utilized grid and boundary layer are shown in Fig. 8. In this case, 9 million elements are used.

3.2.2. Physical discussion

The power coefficient has been selected as a criterion for

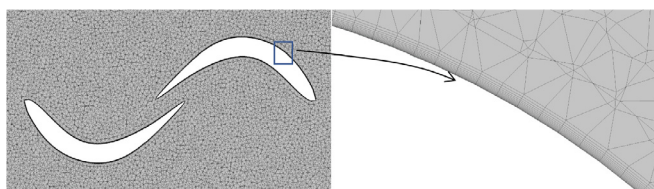


Fig. 8. Details of the mesh: Mesh region in the Sphere (left), boundary layer near the blade (right).

comparison. The power coefficient achieved by the airfoil type turbine using CFD calculations at TSR equal to 0.4 is 0.076 and for the semicircular Savonius is 0.060. Therefore, the optimum airfoil improves the wind turbine performance about 27%. As it was mentioned in the introduction, the improvement of the power coefficient in similar studies depends on the TSR that they chose. Generally, this improvement is between 4% and 30% according to the (Mohamed et al., 2011)–(Marinić-Kragić et al., 2018a), (Mari et al., 2017), (Sharma and Sharma, 2017), (Zhou et al., 2018), (Marinić-Kragić et al., 2018b). However, most of these studies reach an under 20% enhancement on power coefficient which makes the development of this study, acceptable.

The aerodynamic behavior of the wind turbine is a significant part of analyzing procedure. Therefore, the contours of pressure, velocity, and etc. are utilized for comparison. The contours of pressure for the airfoil and the simple semicircular Savonius in TSR equal to 0.4 have been presented in Fig. 9.

The trend for both cases are the same, but the suction area and pressure magnitude of the pressure side of the airfoil type turbine are more extensive than simple type Savonius turbine. Therefore, in the airfoil, the pressure difference causes more force, and in results, an increase in power coefficient occurs.

The static pressure contours for TSR equal to 0.55 and 0.7 are illustrated in Fig. 10. The power coefficient at TSR equal to 0.55 for the optimized airfoil type and simple semicircular Savonius turbine are almost equal. As can be seen, the pressure side in simple semicircular is larger than the airfoil type, but on the other hand, the magnitude of pressure is higher for the airfoil. Furthermore, for the airfoil, the suction side is wider. So the difference between the pressure side and suction side is nearly equal for both cases. As a result, the power coefficients for airfoil and simple semicircular Savonius are very close to each other. For TSR equal to 0.7, the pressure side for the simple semicircular Savonius is significantly increased, and the magnitude of pressure in the suction side is also reduced. Consequently, the power coefficient of semicircular Savonius is being increased considerably in comparison to TSR equal to 0.55. In another hand for the airfoil type, the suction side pressure is being reduced in comparison to TSR equal to 0.55. So the power coefficient for an airfoil is being decreased.

The velocity contours are shown in Fig. 11. It can be seen that the separation is happening faster for the simple semicircular Savonius in both blades.

In Fig. 12 the velocity contours for TSR equal to 0.55 and 0.7 are illustrated. As it is mentioned, the airfoil and semicircular Savonius turbines power coefficients at TSR equal to 0.55 are almost equal. The detachment area for both cases are reduced compared to TSR equal to 0.4, and this reduction in separation area causes more power coefficient for airfoil and semicircular Savonius. The detachment area for an airfoil is larger at TSR equal to 0.7 compare to the airfoil in TSR equal to 0.55, especially for the airfoil facing the free stream. But for semicircular Savonius this area has decreased. For this reason, the power coefficient for an airfoil is being reduced

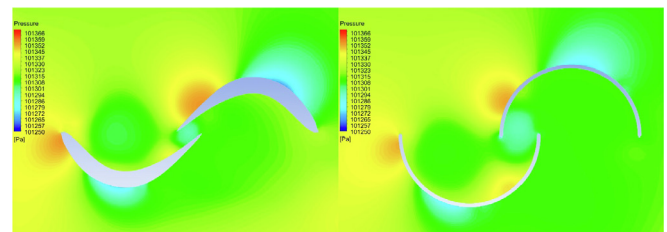


Fig. 9. The contour of static pressure [Pa] for the airfoil and the simple semicircular Savonius (TSR = 0.4).

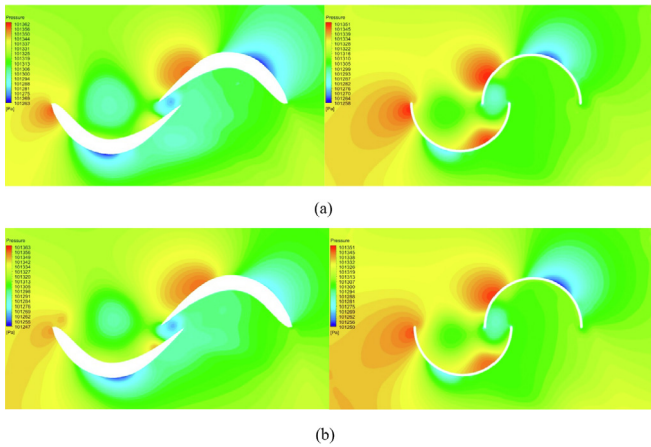


Fig. 10. The contour of static pressure [Pa] for the airfoil and the simple semicircular Savonius (a: TSR = 0.55 and b: TSR = 0.7).

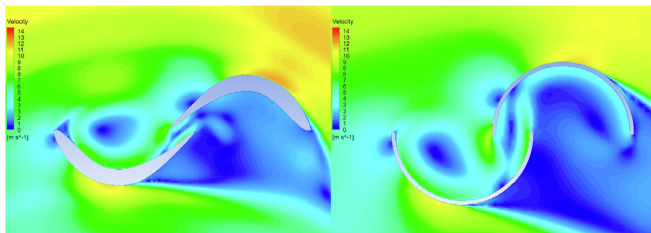


Fig. 11. The contour of velocity for the airfoil and the simple semicircular Savonius (TSR = 0.4).

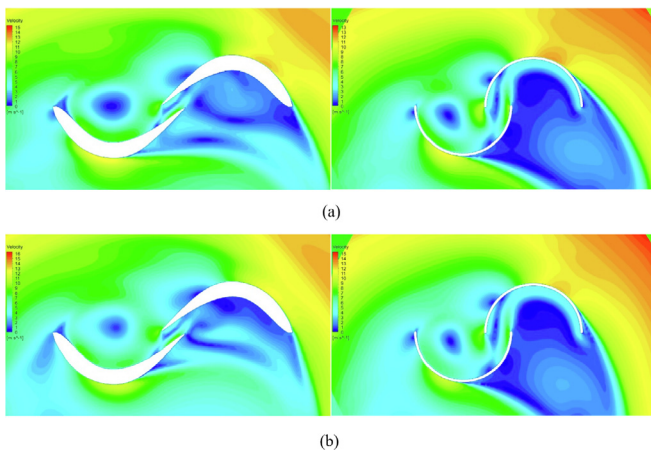


Fig. 12. The contour of velocity for the airfoil and the simple semicircular Savonius (a: TSR = 0.55 and b: TSR = 0.7).

at TSR equal to 0.7. Although, power coefficient for semicircular Savonius has been increased.

Fig. 13 illustrates the turbulent kinetic energy contours. For the detached area in both cases, TKE is increased. In the simple type this area is larger, but in the airfoil, the magnitude of TKE is more significant in some area.

In Fig. 14 the TKE contour for TSR equal to 0.55 and 0.7 is represented. For TSR equal to 0.55 the detachment area for semicircular Savonius is greater than TSR equal to 0.4. The observed increase in the TKE number indicates that the effect of turbulence for higher TSRs in semicircular Savonius is more effective. However,

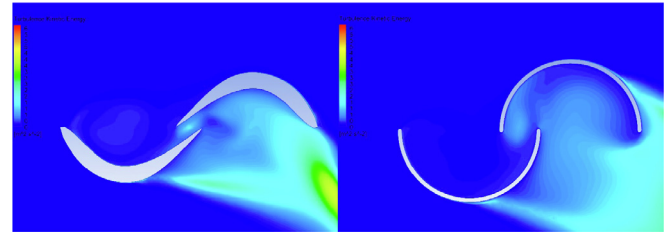
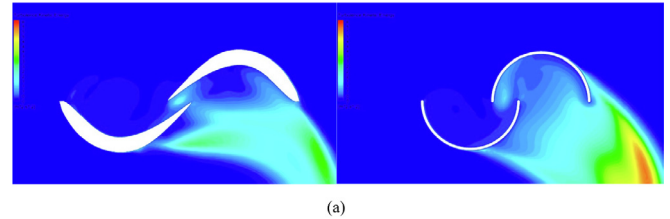
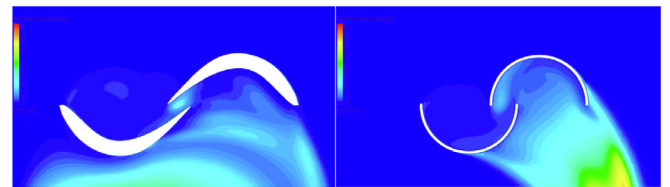


Fig. 13. The contour of Turbulent Kinetic Energy for the airfoil and the simple semicircular Savonius (TSR = 0.4).



(a)



(b)

Fig. 14. The contour of Turbulent Kinetic Energy for the airfoil and the simple semicircular Savonius (a: TSR = 0.55 and b: TSR = 0.7).

this effect for an airfoil in TSR equal to 0.55 is almost constant. For the TSR equal to 0.7, TKE in airfoil type Savonius turbine has experienced a significant increase in number. Also, for semicircular Savonius, the magnitude for TKE number is greater.

4. Conclusion

In this research study, it was aimed to optimize the performance of Savonius wind turbines, by implementing airfoils with optimum coordinates. For this purpose, the Discrete Vortex Method (DVM) code has been developed and was validated using available data in the literature. The developed DVM code was receiving the coordinates of the turbine and was giving the proportional power coefficient. Therefore, it was very suitable to be combined with a Meta heuristic algorithm. Salp Swarm optimization algorithm was selected for this purpose. The reason for selecting this algorithm it was its simplicity and robustness in finding the global optimum solution and its novelty. There are several techniques for optimizing the airfoil coordinates using Meta heuristics algorithms. One of the efficient and accurate methods is the utilization of the Class Shape Transfer (CST) function. In CST a polynomial function is being fitted using the explained procedure in this paper. For the purpose of optimization, the coefficients of the polynomial function are considered as decision variables and the power coefficient is considered as the objective function.

The achievements of this study have been presented in the following lines:

- Developing and validating an unsteady DVM method for the Savonius wind turbine
- Propose a new airfoil-type Savonius wind turbine

- Coupling DVM, CST, and SSA in order to find the optimum parameters of the airfoil
- Presenting an optimum airfoil-type Savonius wind turbine with a 27% increase in the power coefficient
- Verifying the results of optimization with CFD

Many works can be done in the future such as a developing a 3D Discrete Vortex Method solver, representing airfoils with other curve methods such as splines or Bezier curves, finding optimum coordinates at different TSR, defines a new boundary for airfoil coordinates optimization, using CFD as a solver for finding the power coefficient of the wind turbine and experimental study on the optimum Savonius airfoil.

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